

Master's Comprehensive Examination

Part II

Applied (90 minutes)

December 8, 2001

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. Is Longevity the Same for Different Classes of Leaders? The accompanying table shows samples of data from presidents, popes, and British monarchs. The values are the numbers of years the subjects lived after their inauguration, election, or coronation. Test the claim that the three populations are not all the same.

Presidents	26	15	25	1	16	17	0	12	18	21	2	36
Popes	3	18	25	6	15	11	19					
Monarchs	17	13	12	33	10	7	25					

[Based on data from *Computer-Interactive Data Analysis*, by Lunn and McNeil, John Wiley & Sons.]

2. Heights of Presidential Winners and Losers. The accompanying table shows the heights of presidents matched with the heights of the candidates they beat. All heights are in inches, and only the second-place candidates are included. Use the Wilcoxon signed-ranks test to test the claim that both samples come from populations with the same distribution. Is there sufficient evidence to support the claim that height is an advantage to candidates?

Winner	78	66	70	70	74	71.5	73	74
Runner-up	64	71	72	72	68	71	69.5	74

3. In an attempt to study the effectiveness of two corn varieties (V_1 and V_2), two corn-producing counties (C) in Iowa were chosen at random. Four farms (F) were then randomly selected within each county. Seeds from both varieties were sent to these farms and planted in random plots. When harvested, the number of bushels of corn per acre was recorded for each variety on each farm.

- Write a mathematical model for this situation.
- Determine the expected mean squares (EMS) for varieties in this experiment.
- Assuming only counties (C) showed significantly different average yields, and $MS_C = 130$ and $MS_F = 10$ (farms or error), find the percentage of the variance in this experiment that can be attributed to county differences.

4. The tensile strength of Kraft paper (in pounds per square inch) was measured for different percentages of hardwood in the batch of pulp that was used to produce the paper (Joglekar, *et al.*, 1998).

Strength	6.3	11.1	20.0	24.0	26.1	30.0	33.8	34.0	38.1	39.9
Percent hardwood	1.0	1.5	2.0	3.0	4.0	4.5	5.0	5.5	6.0	6.5

- Construct a scatterplot of tensile strength versus hardwood content.
- Does it appear that a straight line fits the data?
- Fit a simple linear regression model to the data. What is R^2 ? Is the F statistic significant?
- Does your answer to part (c) agree with your answer to part (b)?
- Fit a second-degree polynomial to the data. How much improvement is there in R^2 ? Is the F statistic significant? Is this a better model than the simple model in part (c)?

5. In most states, police departments have a certain percentage of unmarked police cars. These are the percentages of marked cars in 28 police departments in Florida:

61	74	49	65
79	64	54	75
56	78	63	74
79	56	69	51
61	61	60	64
51	63	62	62
68	73	56	60

[Source: Bureau of Justice Statistics, *Law Enforcement Management and Administrative Statistics*, 1993, Report NCJ-148825, September 1995, pp.147-148.]

- Construct a histogram of the data. What is the general shape of the histogram?
- Construct a boxplot and a normal probability plot of the data. Are they consistent with the shape given in part (a)? Are there any outliers in the data?
- A criminal justice expert is interested in whether or not the mean for Florida exceeds 60%. Formulate the null and alternative hypotheses for evaluating the question.
- Test the hypothesis stated in part (c). Give the test statistic and its p -value and state your conclusion.